1. Write a program to check if a number is positive, negative, or zero.

In [4]:
```
a=int(input("Please enter a number"))
if a==0:
    print("Number is 0")
elif a>0:
    print("Number is positive")
else:
    print("Number is negative")
```

```
Number is positive
```

2. Check if a given number is even or odd.

In [3]:
```
a=int(input("Enter a number"))
if a%2==0:
    print("The number is Even")
else:
    print("The number is Odd")
```

```
The number is Even
```

3. Determine whether a given year is a leap year or not.

In [13]:
```
year=int(input("Please enter the year"))
if year%4==0:
    if year%100!=0 or year%400==0:
        print("Leap year")
    else:
        print("Not a Leap year")
    # print("Leap year")
else:
    print("Not a Leap year")
```

```
Leap year
```

4. Check if a character entered is a vowel or consonant.

In [2]:
```
a=input("Enter a character").lower()
if a in ["a","e","i","o","u"]:
    print("Vowel")
else:
    print("Consonant")
```

```
Vowel
```

5. Write a program to check if a person is eligible to vote (age ≥ 18).

In [12]:
```
age=int(input("Enter the age to check the eligibilty"))
if age>=18:
    print("Eligible to vote")
else:
    print("Not Eligible to vote")
```

```
Eligible to vote
```

6. Determine the largest of two numbers entered by the user.

In [3]:
```
a=int(input("Enter number 1: "))
b=int(input("Enter number 2: "))
if a>b:
    print("Largest is 1st number: ",a)
elif a<b:
    print("Largest is 2nd number: ",b)
else:
    print("Both are same")
```

```
Largest is 1st number:  5
```

7. Check whether a number is divisible by both 5 and 10.

In [21]:
```
a=int(input("Enter the number to check"))
if a%10==0 and a%5==0:
    print("Number is divisible by both 5 and 10")
else:
    print("not divisible by 5 and 10")
```

```
Number is divisible by both 5 and 10
```

8. Write a program to check if a number is a multiple of 3 or 7.

In [4]:
```
a=int(input("Please enter the number to check"))
if a%3==0 and a%7==0:
    print("The entered number is a multiple of both 3 and 7")
elif a%3==0:
    print("The entered number is a multiple of 3")
elif a%7==0:
    print("The entered number is a multiple of 7")
else:
    print("Not a multiple of 3 or 7")
```

```
The entered number is a multiple of 3
```

9. Check if the input character is an uppercase or lowercase letter.

In [34]:
```
a=input("Enter a character")
if a.isupper()==True:
    print("Upper case")
else:
    print("Lower case")
```

```
Upper case
```

10. Determine whether a given number is a single-digit number or not.

In [ ]:
```
a=int(input("Enter the number"))
if -9<=a<=9:
    print("Single digit")
else:
    print("Double Digit")
```

11. Check if three sides form a valid triangle (sum of two sides > the third side).

In [71]:
```
s1=int(input("enter side 1"))
s2=int(input("enter side 2"))
```

```
s3=int(input("enter side 3"))
if s1+s2>s3:
    print("Valid triangle")
elif s1+s3>s2:
    print("Valid triangle")
elif s2+s3>s1:
    print("Valid triangle")
else:
    print("invalid triangle")
```

```
Valid triangle
```

12. Write a program to find the largest of three numbers.

In [73]:
```
g=int(input("Number 1="))
h=int(input("Number 2="))
i=int(input("Number 3="))
if g>h and g>i:
    print("largest is 1=", g)
elif h>g and h>i:
    print("largest is 2=", h)
elif i>g and i>h:
    print("largest is 3=",i)
else:
    print("all are same")
```

```
largest is 2= 88
```

13. Determine whether a character is an alphabet, a digit, or a special character.

In [8]:
```
a=input("Enter a the character")
if a.isalpha():
    print(a,"is an Alphabet")
elif a.isnumeric():
    print(a,"is a Digit")
else:
    print("special character")
```

```
f is an Alphabet
```

14. Check whether a given number is a prime number.

In [9]:
```
a=int(input("Please enter the number to check="))
if 0<=a<=1:
    print(f"{a} is neither a Prime number or a composite number")
elif a==2:
    print("Prime")
elif a>2:
    if a<=7:
        if a%2>0:
            print("Prime")
        else:
            print("Composite")
    elif a>7:
        if a%2==0 or a%3==0 or a%5==0 or a%7==0 or a%11==0 or a%13==0 or a%17==0:
            print(f"{a} is a Composite Number")
        else:
            print(f"{a} is a Prime Number")
```

```
361 is a Prime Number
```

15. Write a program to calculate the grade based on marks:

In [13]:
```
a=int(input("Please enter the marks"))
if 90<=a<=100:
    print("A")
elif 80<=a<=89:
    print("B")
elif 70<=a<=79:
    print("C")
elif 60<=a<=69:
    print("D")
else:
    print("F")
```

```
C
```

16. Write a program to check if a given temperature is hot (> 35°C), moderate (15°Cto 35°C), or cold (< 15°C).

In [60]:
```
temp=int(input("Please enter the temparature is Celsius"))
if temp>35:
    print("hot")
elif 15<=temp<=35:
    print("moderate")
else:
    print("cold")
```

```
cold
```

17. Determine whether a given number is a palindrome (reads the same backward and forward).

In [15]:
```
a=input("Enter the value")
if a==a[::-1]:
    print("Palindrome")
else:
    print("Not a Palindrome")
```

```
Not a Palindrome
```

18. Write a program to check if a person is eligible for a senior citizen discount (age≥ 60).

In [52]:
```
age=int(input("Enter the age"))
if age>=60:
    print("The person is eligible for a senior citizen discount")
else:
    print("The person is not eligible for a senior citizen discount")
```

```
The person is not eligible for a senior citizen discount
```

19. Check if the sum of two given numbers is greater than 100.

In [38]:
```
a=int(input("Enter 1st number"))
b=int(input("Enter 2nd number"))
if a+b>100:
    print("Addition of the given numbers exceeds 100")
```

```
else:
    print("not")
```

```
Addition of the given numbers exceeds 100
```

20. Write a program to classify a given day of the week as a weekday or weekend.

In [50]:
```
weekend=["saturday","sunday"]
weekday=["monday","tuesday","wednesday","thursday","friday"]
day=input("Enter a day").lower()
if day in weekend:
    print("Entered day is a Weekend:", day.capitalize())
elif day in weekday:
    print("Entered day is a Weekday:", day.capitalize())
else:
    print("Invalid expressions, run the code again")
```

```
Entered day is a Weekday: Friday
```